

week-4-lesson-2

Example-1

Write a program that shows the result a student qualifies for based on the CSCI-151 course GPA. GPA to pass the course = 2.0 and gpa to pass to the SCAI major = 3.0.

Example-2

Show the opportunity a student has before the class at 10 depending on the wake up time. The messages are: “can do some morning exercise”, “prepare for the classes”, “have breakfast” and “Hurry up! You are late”.

Example-3

[Collatz Conjecture Step](#) Tracker (Conditional Operator + while + break)

Track the step sequence for a starting positive integer $n = 13$, printing each value until the sequence converges to 1. Loop Structure: Use a while loop running as long as n is not equal to 1. Branchless Step Formula: Inside the loop, update the value of n using a single ternary conditional operator expression: if n is even, divide by 2; otherwise, multiply by 3 and add 1. Infinite Loop Prevention: Incorporate a step counter. If the number of steps exceeds 100, execute a break statement to force-terminate the loop as a safety mechanism.

Example-4

Print the floor numbers from 1 to 10 in a separate line. Print 3A when it reaches number 4. Use continue statement.